



University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February 2025

IT2114 Data Structures (Theory)

Answer any Four Questions

Time Allowed : Two hours

30

a way to organizing and storing data

1. (a) Write the significance of *data structures* in computer science. [10%]
- (b) Write down *two* examples for each *linear* and *non-linear* data structure and explaining their characteristics and use cases. [15%]
- (c) Define each of the following terms:
 - i. *Performance analysis*
 - ii. *Space complexity*
 - iii. *Time complexity*

function call itself
base
No loop

base X
loop
- (d) State four differences between the *recursive* and *iterative* algorithms. [15%]
- (e) Write an algorithm to generate the *Fibonacci sequence* using the *recursive method*. [25%]

[This question is continued on the next page]

(f) Determine the *time complexity* of the following code segment:

```

Algorithm_For_Problem(S, P, A, B, G, H)
{
    input:
    S = Matrix of size AxH
    P = Matrix of size GxH
    output:
    Q = The matrix product of S and P
    if B = G:
        Q := make a new matrix of size AxH
        for m := 0 to A - 1:
            for r := 0 to H - 1:
                Q[m, r] := 0
                for k := 0 to G - 1:
                    Q[m, r] = Q[m, r] + S[m, k] * P[k, r]
                return Q
            else:
                return "Incompatible dimensions"
        }

```

[20%]

2. (a) Explain both *Stack* and *Queue* data structures with an aid of an example. [15%]

(b) Explain the need for a *circular queue* and describe a scenario where it can be used. [15%]

(c) Write a Java program to perform the operations, *enqueue* and *dequeue*, in a circular queue. [20%]

(d) Convert the following *infix expression* into a *postfix notation* using a *Stack* data structure:

$((a+b) \times (c - (d/e) + f)) - (g / (h+i)) + ((j \times k) - (l+m/n))$

[25%]

(e) Write a Java program to implement the Selection sort algorithm. [15%]

(f) Sort the following array of elements using the algorithm in part(e):

42, 7, 19, 3, 56, 23, 11

[10%]

3. (a) Write the primary limitation of *array* data structures, and discuss briefly how this limitation is resolved by utilizing *linked list* data structures. [10%]
- (b) Explain the differences between a *singly linked list* and a *doubly linked list*. [15%]
- (c) Figure 1 shows a *doubly linked list* consisting of five nodes; the first node is referred to as the head node, while the last node is known as the tail node. Each node contains a data field to store elements, a *next* field that points to the subsequent node in the list, and a *prev* field that points to the previous node.

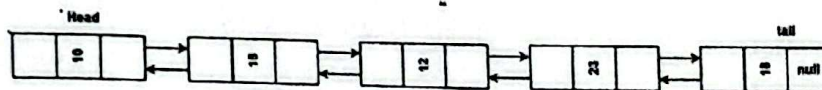


Figure 1: A doubly linked list

For each of the following codes segments A, B, C, and D given below, draw the corresponding conceptual linked list.

i. Code A

```

new_node.value = 08;
new_node.next = head;
head.prev = newnode;
head = newnode;
  
```

[05%]

ii. Code B

```

current_node = head;
while(current_node != null && !current_node.data.equals(12))
{
    current_node = current_node.next;
}
current_node.prev.next = current_node.next;
  
```

[05%]

iii. Code C

```

current_node = head;
for(int i=0; i<3; i++)
  
```

[This question is continued on the next page]

```

{
    current_node=current_node.next;
}
current_node.next=null;

```

[05%]

iv. Code D

```
head.next.next.prev.data=tail.data
```

[05%]

(d) Consider a Class named Chain that includes two data members: head and size. The data type of the head node is ChainNode. The ChainNode class is designed to create nodes that contain an element and a next field. The data type of the next field is ChainNode, while the element can be an object. The nodes in the chain are connected to one another using the next field.

i. Create a Class for a chainNode.

[10%]

ii. Write a Java Class for a chain to include the following methods:

A. Insert a node at a specific index.

[15%]

B. Obtain the index of the specified data.

[15%]

C. Remove a node that contains a specified value.

[15%]

4. (a) Define a *graph* using appropriate notation in the context of data structure with an aid of an example and its description

[15%]

(b) Describe each of the following graphs with the aid of an example:

i. *Simple graph*

ii. *Complete graph*

iii. *Directed graph*

iv. *Undirected graph*

[20%]

(c) Draw the *adjacency matrix* and *adjacency list* for the graph shown in Figure 2 .

[20%]

(d) Compare *breadth-first search method* and *depth-first search method*.

[15%]

[This question is continued on the next page]

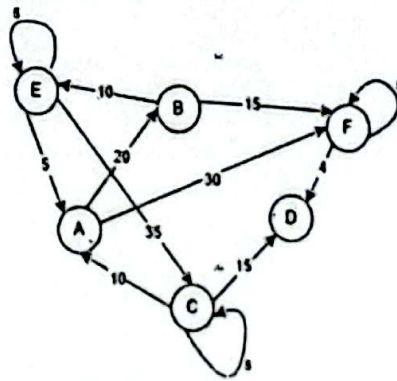


Figure 2: Graph for adjacency matrix and list

- (e) For each of the following search methods, determine the sequence of vertex visits for the graph in Figure 3, starting from vertex v_0 .

- i. *Breadth first search*
- ii. *Depth first search*

[30%]

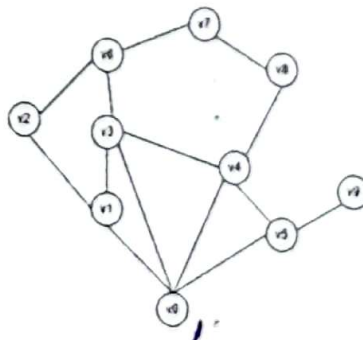


Figure 3: Graph for BFS and DFS

5. (a) Refer to the *Tree T* shown in Figure 4 and identify each of the following components:
- i. *Root node*
 - ii. *Siblings of the Node F*

[This question is continued on the next page]

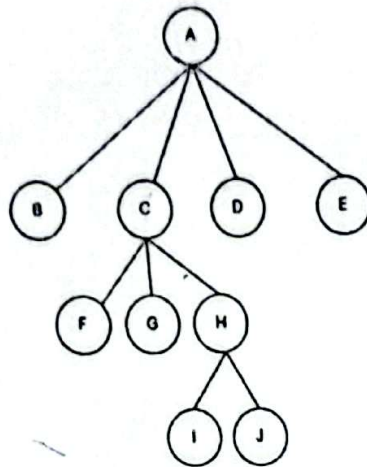


Figure 4: Tree T

iii. *Internal nodes*

iv. *Height of the tree*

v. *A sub-tree*

[15%]

(b) Define the properties of a *binary search tree (BST)* and illustrate it with the aid of a tree diagram.

[15%]

(c) Construct a binary search tree by entering all the following data elements sequentially from left to right: 50, 30, 70, 20, 40, 25, 35, 55, 65, 85

[15%]

(d) Perform each of the following traversal methods on the constructed BST in part (c) and write down the order of visited nodes:

i) *Pre-order* ii) *Post-order* iii) *In-order*

[20%]

(e) Draw two appropriate possible trees after removing the node 70.

[15%]

(f) Construct a binary tree using the pre-order and in-order traversal techniques with the following visited nodes:

Pre-order: P A S T Q E D X M R C F

In-order: T S Q A E D P M X C R F

[20%]



University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February - 2025

IT2114 Data Structures (Practical)

Answer All Questions

Time Allowed : Three hours

1. (a) Write a Java Class to implement a Stack Abstract Data Type using an array with the following methods: *push()*, *pop()*, *isEmpty()*, and *isFull()*.

- (b) Modify your program in part(a) to evaluate a post-fix expression given by a user.

- (c) Test your program in part (b) for the following post-fix expression:

5 6 2 + * 9 -

[20%]

2. (a) Write a Java program to implement a binary search tree with the following methods:

insert(), *search()*, *postOrder()*, *inOrder()*, and *preOrder()*.

- (b) Insert the following nine values into the binary search tree:

50, 30, 70, 20, 40, 25, 35, 65, and 85.

- (c) Display the values of nodes in the binary search tree for each of the following traversals: in-order, pre-order, and post-order.

- (d) Check whether the node with the value 25 is found in the binary search tree. If it is found, display the message as "Node 25 is found"; otherwise, display "Node 25 is not found."

[20%]

3. An admission branch of a University decided to manage student registrations for three courses to maintain the students for different stages of waiting list provided by the UGC after the enrollment of the initial registration for each of the courses.

(a) You are required to create a Java Class with the name "UniversityRegistration" to implement a Circular Queue using an array with the following methods: *enqueue()*, *dequeue()*, *isEmpty()*, *isFull()*, and *printQueue()*.

(b) Consider the following:

i. A registered student in a University System will have a unique registration number. The first three characters of a registration number indicate the course a student pursues. Assume that if a registration number starts with 'C', the pursuing course is Computer Science; if it starts with 'S', the pursuing course is Statistics; or if it starts with 'D', the pursuing course is Data Science.

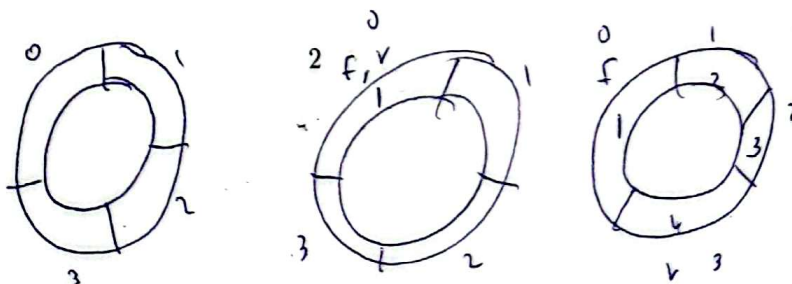
ii. Each course is allowed to register up to 10 students. If a course reaches its maximum capacity, additional students can be added to a waiting list, which also has a maximum capacity of 5 students.

iii. The following 44 registration IDs were enrolled in the given order to assign appropriate courses to the students:

Cabs01, abs02, abs03, abs04, abs05, abs06, abs07, abs08, abs09, abs10, abs11, abs12, abs13, abs14, abs15, Cabs16, sdf01, sdf02, sdf03, sdf04, sdf05, sdf06, sdf07, sdf08, sdf09, sdf10, sdf11, sdf12, sdf13, sdf14, sdf15, klm01, klm02, klm03, klm04, klm05, klm06, klm07, klm08, klm09, klm10, klm11, klm12, and klm13.

(c) Using the program implemented in part (a), create queues for each of the following courses: course_A, course_B, and course_C, along with their corresponding waiting list as: waiting_list_A, waiting_list_B, and waiting_list_C respectively. After the completion of the students registration process, print the lists of students for each of the Course such as Course A, Course B, and Course C.

[This question is continued on the next page]

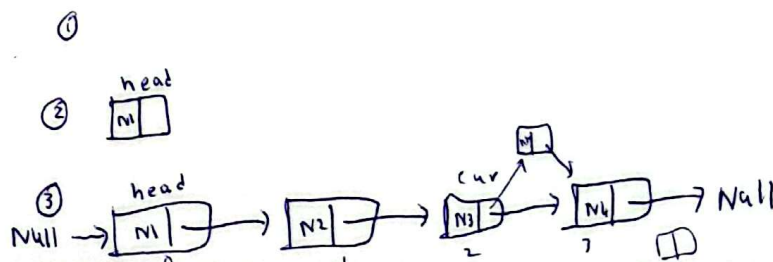


- i. Print the waiting lists for each of the courses such as `waiting_list_A`, `waiting_list_B`, and `waiting_list_C`.
- ii. Suppose that the first student in Course A has dropped out; then transfer the first student from the waiting list of Course A to the queue Course A.
- iii. Print the updated registration list and waiting list for Course A. [30%]

4. Write a Java program for a Library Book Management System (LBMS) using a **Singly linked list**. The aim is to create a program to manage necessary operations such as *add*, *delete* and *search* using a singly linked list. Consider the following functionalities in order to perform each of the necessary operations:

- (a) When a new book arrives at the library, it can be added to the LBMS, which includes the associated book ID, book name, and author's name.
- (b) Suppose that if a book is damaged, it can be removed from the list.
- (c) Write a method `"insert_front"` to add books to the list; each book should have a unique book ID, a name, and an author's name. *remove at index*
- (d) When removing a book, get a prompt from the user and remove the corresponding book based on the book ID if it is available; if not, display an appropriate error message.
- (e) When searching for a book, the user should enter the book ID. Display the book's details, such as the book name and author's name; if it is valid, if not, display a message, "it is an invalid book ID".
- (f) Display all the books in the library along with their details, book ID, book Name, and author's name in the current order. *Display list*
- (g) Test the *add* method by getting the request from the user to enter a valid index, and with its ID, name, and author name, add it to the list in proper. *insert at Index*
- (h) Test the delete method by getting it from the user and displaying the final list. [30%]

3





University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February 2025

IT2122 Software Engineering

Answer Four Questions Only

Time : Two hours

1. Software drives modern innovation, transforming how we connect, work, and solve challenges.

(a) Describe what is meant by software engineering in your own words. [15%]

(b) Compare and contrast a web-based software system and a desktop-based software system. *activity, maintainability, integrity, reliability & accessibility, secure* [20%]

(c) Briefly describe four important attributes of good software. Explain why these attributes are critical in software development. [20%]

(d) Explain the difference between software engineering and systems engineering. Give examples to illustrate your points. [15%]

(e) State two special characteristics and give one relevant example for each of the following types of software products:

i. Embedded software

ii. Real-time software - *air traffic control*

iii. System software - *windows* [15%]

(f) Identify three major challenges faced by software engineers in the current world IT industry. [15%]

2. The Software Process is a structured set of activities required to develop a software system.

(a) List the major software development activities. [15%]

(b) Describe what a software process model is and briefly explain the purpose it serves in software development. [20%]
set of activities to do in

(c) Many software process models are currently used by developers, and their selection depends on various factors. *ex. E*

i. Describe the incremental development model with the aid of a diagram. [15%]

ii. Identify two advantages of the incremental development model over the waterfall model. [10%]

iii. List the four main task regions of the spiral model. [10%]

(d) Agile is a popular process model known for its unique values and principles.

i. Describe the Agile model and briefly explain how it works. [15%]

ii. Outline the key steps involved in the Scrum process. [15%]
Product backlog creation, Sprint planning, Sprint backlog creation, Product Catalog, Sprint, Implementation, Testing and Implementation, Catalog, Sprint review and feedback

3. (a) Describe the importance of requirement engineering in the software development process. [15%]

(b) List four requirement elicitation methods. *interviews, talking with stakeholders* [10%]

(c) Explain the differences between user requirements and system requirements, providing an example to illustrate your explanation. [20%]

(d) Consider the following description of a software system:

A University Course Management System (UCMS) is designed to manage course-related activities for students and staff. The system maintains information about courses, students, instructors, and the schedule of classes. The following are some key features of the system:

- Instructors can create and manage course details, including syllabus, schedules, and grading criteria.

[Question 3 continues on next page.]

- Students can register for courses, view their schedules, and access course materials. The system automatically checks prerequisites when students register for a course. The system generates reports on student performance, course enrollments, and instructor workload. Notifications are sent to students and instructors for changes in schedules or announcements.

- Identify the stakeholders of the above system. [10%]
- List five functional requirements of the above system. [15%]
- Identify three non-functional requirements of the above system and justify why each is important for the system. [15%]
- Explain the interaction scenarios between the system and stakeholders. [15%]

4. UML is a pictorial language used to make software blue prints.

- Describe the major components of a UML use case diagram. [20%]

- Consider the following required functionality of a print-on-demand web service:

The print-on-demand service allows users to create custom print orders for posters, flyers, or books. Users can select poster, flyer, or book, specify the quantity, and choose the paper type. For book orders, users must also select between hard or soft covers and upload a PDF content. To place an order, users must log in or create an account, which requires a username, password, and valid payment details (e.g., credit card). The system ensures all necessary information for the order and the user's account is complete. Once all information is validated, the order is submitted, and payment details are sent to the bank for authorization. Upon successful payment approval, the order is confirmed. A printing agent reviews the uploaded PDF files for books or any other content requirements and manages the printing process to ensure the order is fulfilled accurately.

Draw a UML Use Case Diagram for the print-on-demand service. [30%]

- Describe the basic elements and their notations of a UML Sequence Diagram. [20%]

[Question 4 continues on next page.]

(d) A **Library Management System** is designed to manage books, members, and library staff. The system allows members to borrow and return books, while staff manage the inventory and oversee member activities. The **Book** contains attributes: title, author, ISBN, publicationYear, and availabilityStatus, along with methods: checkAvailability() and updateStatus(). The **Member** has attributes: memberID, name, email, phone#, and borrowedBooks, with methods: borrowBook() and returnBook(). The **Staff** includes attributes: staffID, name, position, and salary, with methods: addBook(), removeBook(), and updateInventory(). The **Library** class contains attributes: libraryName, and address, along with methods: registerMember() and manageBooks(). The library has multiple books, members, and staff. Members borrow and return books (one member can borrow many books, but a book can only be borrowed by one member at a time) and staff communicate members and manage books.

Draw a UML class diagram for the Library Management System described above.

State any assumptions that you make.

[30%]

5. (a) Compare and contrast verification and validation in the context of software testing, with the aid of examples.

[15%]

(b) Define the terms test case and test set with the aid of examples.

[10%]

(c) Briefly describe the different stages of software testing.

[20%]

(d) Compare and contrast black box testing and white box testing in software development. Give examples where each testing methodology is suitable.

[20%]

(e) List the four main types of software maintenance activities.

[15%]

(f) An e-commerce company has a website that allows customers to browse products, make purchases, and track their orders. After several months of operation, the company receives customer feedback regarding issues with the product search functionality and the checkout process. Additionally, the website's performance has been slowing down during peak shopping hours.

Explain how the maintenance activities listed in part (e) would be applied to this software system. State any assumptions that you make.

[20%]



IT17099

University of Vavuniya

Second Examination in Information Technology - 2023

^{First}
~~Second~~ Semester - January/February 2025

IT2143 Visual Programming (Practical)

Answer All Questions

Time Allowed : Three hours

Instructions:

- All the necessary images are given in the folder IT2143-Resources on your PC's Desktop.
- You should use C # programming language and Visual Studio IDE to develop the application.
- Create a new folder with your index number on your PC's Desktop and use it to save your project files.

1. The "Your Life Hotel" has decided to develop a Windows-based application consisting of multiple windows forms to Manage bookings in their hotel. The initial Windows form of the application should be a login form (refer to Figure 1), with the username 'user' and the password '1234'. [15%]
2. After a successful login, the application should display a dashboard with a button labeled 'Book a Room' (refer to Figure 2). [10%]

3. When you click the 'Book a Room' button on the dashboard, a Windows form titled 'Book Now' should be displayed (refer to Figure 3). The functionalities of the 'Book Now' form are detailed below:

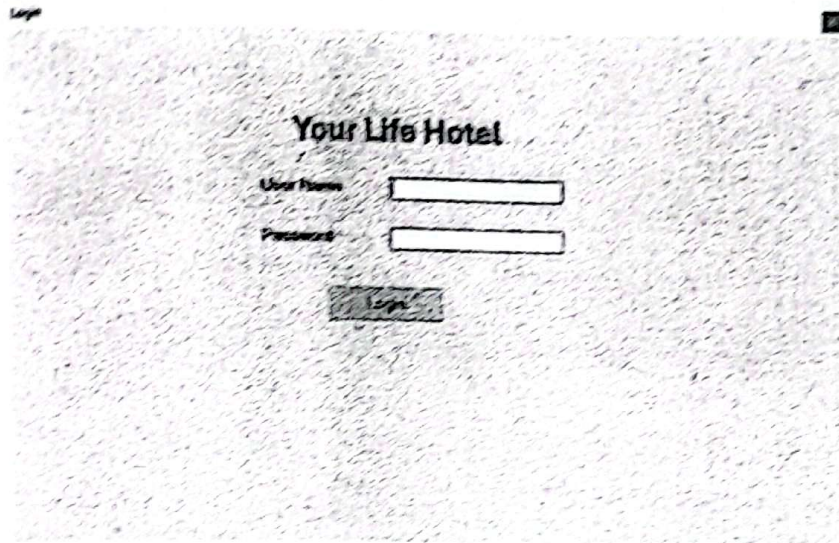
- **Guest Details:** Provide text boxes to enter guest names and contact information.
- **Room Type Selection:** Customers should select the "Room Type" from the following categories:
 - Standard: Rs. 5000
 - Deluxe: Rs. 7500
 - Suite: Rs. 12,000
- **Customization Options:** Allow customers to add facilities:
 - Breakfast: Rs. 500
 - Airport Pickup: Rs. 1,500
 - Extra Bed: Rs. 2,000
- **Stay Duration:** Customers should select the number of nights for the stay.
- **Clear Button:** Clear all fields to start a new booking.
- **Additional Costs:** The final price should include:
 - 10% service charge.
 - Taxes (15% of the total cost).
- **Calculate Cost:** Calculate the total cost based on the selected options.
- **Booking Confirmation:** After clicking the *Book Now* button, display a message box to confirm the booking with the total amount (refer to Figure 3). [50%]

4. After clicking Book Now, show a new form displaying the following booking details: (refer Figure 4). [15%]

- Guest Name
- Contact Information
- Room Type

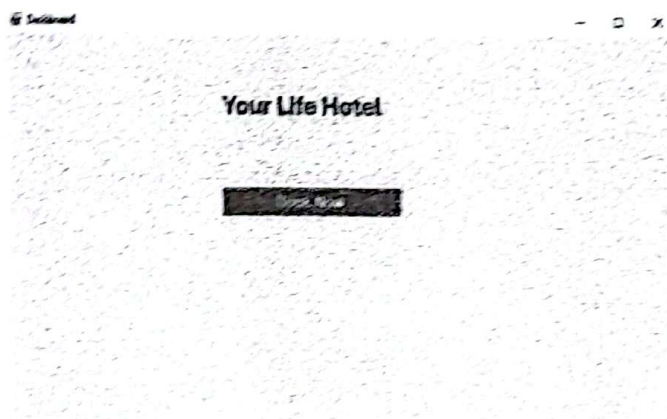
- Facilities Added
- Check-in Date
- Number of Guests
- Number of Nights
- Total Cost (including service charges and taxes)

5. Provide a button labelled 'Got it', which closes the summary form when clicked. [10%]



The screenshot shows a web browser window with a login form. The form has a title "Your Life Hotel" at the top. Below the title, there are two input fields: "User Name" and "Password". Below these fields is a "Login" button. The browser's address bar shows "http://localhost:8080/".

Figure 1: Login form



The screenshot shows a web browser window with a dashboard. The dashboard has a title "Your Life Hotel" at the top. Below the title, there is a "Logout" button. The browser's address bar shows "http://localhost:8080/".

Figure 2: Dashboard

Book a Room

Guest Name: Check-in Date:

Contact No: Number of nights:

Room type: Facilities:

- ☒ Breakfast
- ☒ Airport Pickup
- ☒ Extra Bed

No. of Guests:

Booking Total

Room Price: Rs. 5000
 Additional Services: Rs. 4000
 Service Charge (10%): Rs. 900.00
 Taxes (15%): Rs. 1350.00
 Total Cost: Rs. 11250.00

Figure 3: Sample Booking Details and Confirmation Message

Summary

Your Booking

Name: Anne
 Contact No: 0786672619
 Room type: Standard
 No. of guest: 1
 Facilities:
 Check-in Date: 2025-01-17
 Number of nights: 1
 Total cost: 11250.00

Rich text box

2/27/2025

Figure 4: Sample Booking Summary



University of Vavuniya

Second Examination in Information Technology - 2023

^{first}
~~Second~~ Semester - January/February 2025

IT2143 Visual Programming (Theory)

Answer any two Questions

Time Allowed : One hour

1. (a) Define what is meant by 'Visual Programming', and explain how it is different from traditional programming. [15%]
 - (b) i. Briefly explain the Rapid Application Development (RAD) methodology. [15%]
ii. Develop a step-by-step process to design and implement a simple Library Management System using RAD. [25%]
 - (c) Describe two challenges in RAD-based projects and propose solutions to overcome them. [20%]
 - (d) State three differences between process-oriented and event-driven programming. [15%]
 - (e) Discuss how process-oriented development ensures better organization in software projects. [10%]
-
2. (a) Discuss three essential components of a well-designed GUI with examples. [15%]
 - (b) Draw a simple GUI layout for a Task Management Application and label its elements. [25%]

[This question is continued on the next page]

- (c) Describe three common mistakes in GUI design and their impact on usability. [25%] ✓
- (d) Explain how does quality management ensure a reliable visual programming application with the aid of an example. [20%]
- (e) Discuss the actions you would take to improve the user experience (UX) of a GUI-based application. [15%] ✓
3. (a) i. Describe the process of creating an executable file for a visual programming application. [20%] ✓
 ii. Why is this step important for software deployment? [10%]
- (b) Write a pseudo-code or algorithm for an event-driven program that handles a Login Button click in a login form. [20%] ✓
- (c) List down and explain three mistakes commonly made in event-driven programming. [15%] ✓
- (d) Explain how event-driven programming enhances the interactivity of a Simple Calculator application with the aid of an example. [20%] ✓
- (e) Discuss the challenges of debugging event-driven programs and suggest tools or techniques to address them. [15%]



University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February 2025

IT2153 Computer Graphics (Theory)

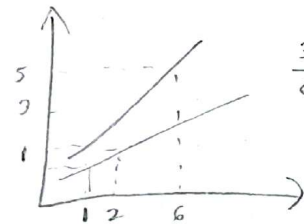
Answer any Four Questions Only

Time Allowed: Two Hours

1. (a) Describe what is meant by *scan conversion* in Computer Graphics. [10%]

(b) Briefly describe each of the following anti-aliasing techniques used to address the aliasing problems:

- i. Super sampling.
- ii. Area sampling,
- iii. Pixel phasing
- iv. Masking.



$$\frac{3-1}{6-1} = \frac{2}{5} = 0.4$$

$$\frac{3-1}{6-2} = \frac{2}{4} = 0.5$$

[20%]

(c) Write *DDA* line drawing algorithm to draw lines when $|m| \leq 1$. [20%]

(d) Derive the necessary equations for the *Bresenham* line drawing algorithm for a positive slope less than one. [20%]

(e) Determine the coordinate points of a line between (10,20) and (18,25) using the *Bresenham* line drawing algorithm. [30%]

2. (a) Write down recursive filling algorithms for the *interior defined* and *boundary defined* regions using 4-connected neighbouring pixels.

(b) Write an algorithm or procedure to draw a circle using the *Bresenham's* circle drawing algorithm. [30%]

(c) Determine the pixel positions along the circle octant of radius $r = 10$ from $x = 0$ to $x \leq y$ centered at the origin using the *Bresenham's* circle drawing algorithm. [30%]

(d) Briefly describe what *ray tracing* is and explain how it differs from rasterization in rendering 3D graphics. [15%]

3. (a) Define the *out-codes* for a typical rectangular clipping area to clip lines using the *Cohen-Sutherland* algorithm. [20%]

(b) Describe the *Cohen-Sutherland* clipping algorithm by defining the necessary equations to determine the coordinates of the intersection points of a line with the horizontal and vertical boundaries of the clipping window. [30%]

(c) Apply the *Liang-Barsky* clipping algorithm to clip a line XY , where $X(-15, -30)$ and $Y(30, 60)$ by determining its intersection points with a rectangular clipping window $KLMN$ defined by $K(0, 0)$ and $M(15, 15)$. [30%]

(d) Briefly describe each of the following algorithms for hidden surface removal in Computer Graphics:

- Depth-buffer (or *Z-buffer*) algorithm.
- Painter's algorithm.

[20%]

4. (a) Derive equations for window to viewport transformation of a point (w_x, w_y) in the window maps to the point (v_x, v_y) in the viewport in such a way to preserve proportion. Clearly state all assumptions made if any. [20%]

- (b) Derive a 3×3 homogeneous coordinate matrix for each of the following transformations:

- i. Translating a point $P(x, y)$ through $(dx1, dy1)$ followed by a translation of it by $(dx2, dy2)$ about the origin. [10%]

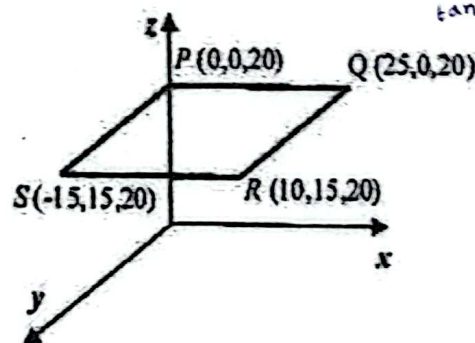
- ii. Rotating an object anti-clockwise through the angle θ about a point $P(x, y)$. [15%]

- (c) Consider an object $ABCDEF$ with the coordinates $A(2, -5)$, $B(7, -5)$, $C(9, -3)$, $D(9, 0)$, $E(4, 0)$, and $F(4, -3)$.

Find the coordinates of the object after each of the following transformations in the homogeneous coordinate system:

- i. Scale the initial object to half of its size about the point F .
ii. Rotate the initial object by 60° anti-clockwise about the point C .

- (d) Consider the following three dimensional object $PQRS$:



Find the coordinates of the object after each of the following transformations in the homogeneous coordinate system:

- i. Reflect the initial object with respect to the xy plane.
ii. Rotate the initial object by 90° about the z axis.

[25%]

5. (a) Describe each of the following parallel projections with the aid of a diagram:

i. Orthographic

ii. Oblique

[20%]

(b) Derive a homogeneous transformation matrix for perspective projection of a point

$P(x, y, z)$ on the XY view plane, if the centre of the projection is $(0, 0, -d)$. [20%]

(c) Suppose that the coordinates of a three dimensional cube $ABCDEFGH$ are $A(0, 0, 0)$, $B(10, 0, 0)$, $C(10, 10, 0)$, $D(0, 10, 0)$, $E(0, 10, 10)$, $F(0, 0, 10)$, $G(10, 0, 10)$, and $H(10, 10, 10)$.

Find the coordinates of the projected image on the XY view plane using the projection matrix obtained in part 5(b), if the centre of the projection is $(0, 0, -6)$.

[20%]

(d) A Bézier curve is a weighted sum of $n + 1$ control points $P_0, P_1, \dots, P_n$ where the weights are the Bernstein's polynomials and given by

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i, \quad 0 \leq t \leq 1$$

i. Derive the cubic Bézier curve equation and find the coordinate functions $x(t)$, $y(t)$, and $z(t)$ of the Bézier curve with the control points $P_0(0, 0, 0)$, $P_1(1, 0, 0)$, $P_2(0, 1, 0)$, and $P_3(0, 0, 1)$. [20%]

ii. Derive the quadratic Bézier curve equation using De Casteljau's algorithm and determine the algebraic expression of the quadratic Bézier curve with the control points $P_0(0, 0)$, $P_1(2, 3)$, and $P_2(5, -1)$. [20%]



IT17099

University of Vavuniya
Second Examination in Information Technology - 2023
First Semester - January/February 2025
IT2153 Computer Graphics (Practical)
Answer All Questions

Time Allowed: Two Hours

Instructions:

- Create a folder on the desktop with your index number and save all your files in it.
- You may see the output images in the given *Resources_IT2153_P* folder stored in the shared folder to view the colour representations.
- It is your responsibility to save your programs frequently.

1. Write a Java program to perform each of the following tasks:

(a) Implement the DDA or Bresenham's line drawing algorithm to draw a straight line. [15%]

(b) Construct a blue triangle ABC where A(200,400), as shown in Figure 01 with the given attributes using the line drawing algorithm in part (a) above while keeping the Window size as 800×800 pixels (You may refer the Resources folder to see the colour representation of the object.):

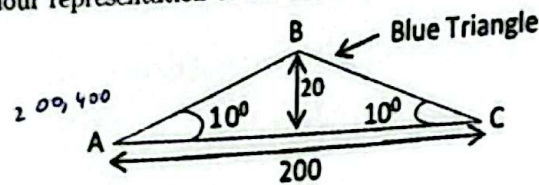


Figure 01

(c) Construct an object in blue as shown in Figure 02 using the object in part (b) above and its reflection transformation. Ensure that the reflected triangle is constructed accurately to form the complete compound object as depicted in Figure 02. Note that you may derive the composite transformation matrix to accomplish the task: [10%]

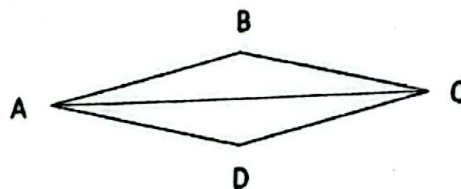
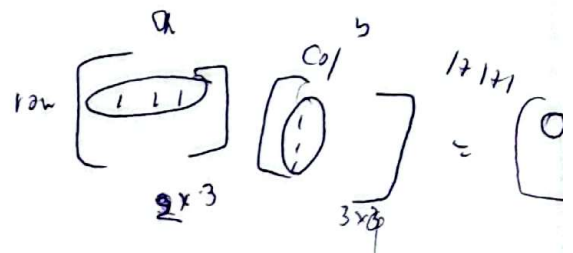


Figure 02

(d) Construct another compound object in blue as shown in Figure 03 by applying the rotation transformation about the centre A. Apply the transformation on the object ABCD in Figure 02 above and rotate it by 45° successively to generate the compound object as in Figure 03. Ensure that each of the rotated instances of the object ABCD should be displayed correctly to form the compound object: [10%]

[This question is continued on the next page.]

2



$$x' = \frac{ax + by}{ay' + bx'}$$

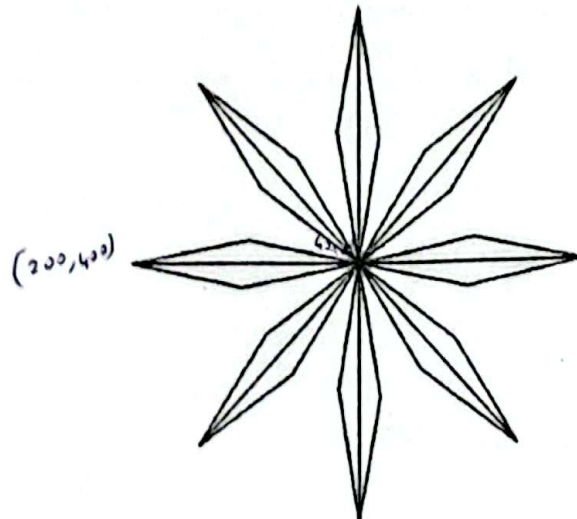


Figure 03

- (e) Apply a scaling transformation by a factor of 0.5 to the compound object in Figure 03 above to construct the compound object as shown in Figure 04 in blue and red. You may refer the *Resources* folder to see the colour representations of the object. Ensure that the scaled object retains its proportions and is correctly positioned relative to the centre: [10%]

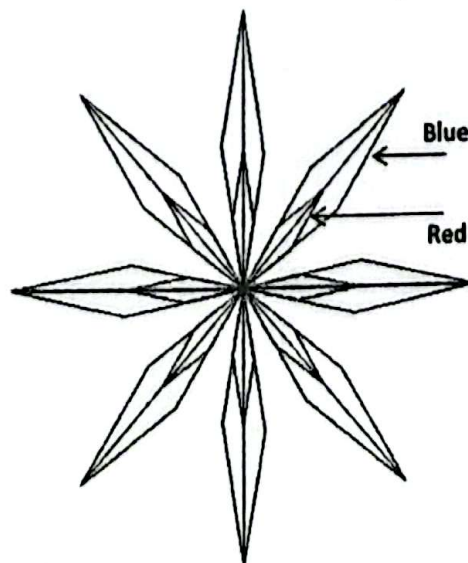


Figure 04

[10%]

- (f) Implement *Bresenham's* circle drawing algorithm to draw a circle. [15%]
- (g) Using the object in Figure 04 above and the algorithm in part (f) above, construct an object as shown in Figure 05 given below:

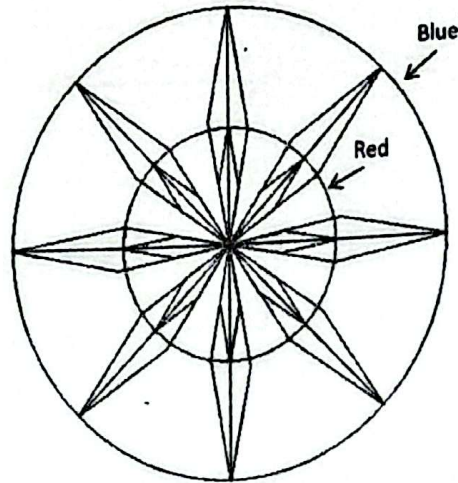


Figure 05

- (h) Construct another object as shown in Figure 06 using appropriate transformation(s). [15%]

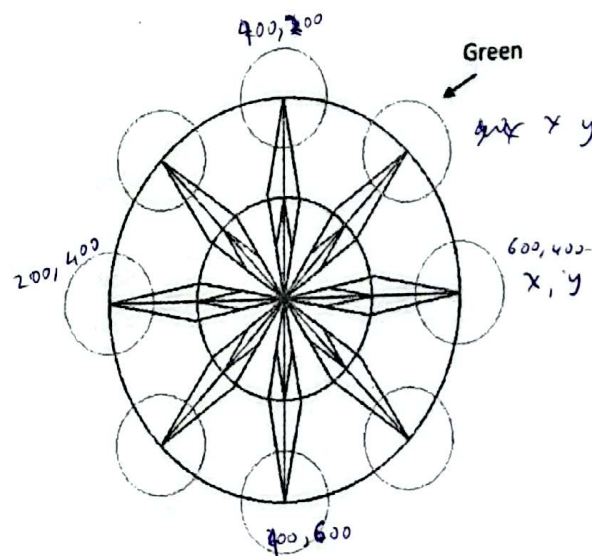


Figure 06



University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February 2025

IT2133 Advanced Web Programming (Practical)

Answer All Questions

Time Allowed: Two Hours

Instructions:

- Source files, MySQL dump file for Database and layout Figures of the expected design (given in Reference Images folder) are stored in the Resources folder.
 - Conceptual design of web page navigation is given in the Figure 10 and stored in Reference Images Folder.
 - Create a folder on your Desktop, name it with your index number and save all files into that folder. Use XAMPP to run the php applications and PhpMyAdmin to run the database required.
1. The administration of the University of Vavuniya has decided to implement a QR system to manage all inventory items. You are tasked with developing a web-based Inventory Management System for the university staff using PHP, HTML, and CSS, JavaScript considering the provided requirements.
- (a) Develop a Configuration page that encompasses variables for database connection details (server name, username, password, and database name). This page serves the purpose of establishing a connection between the application and the database. Integrate this configuration file into various PHP files to facilitate the establishment of a database connection.

(b) Create an **Index** page as shown in *Figure 1* which has two links for **Login** and **Register** pages.

(c) Create a *Registration page* as shown in *Figure 2* which authorizes the staff to create an account. The form should include the following fields:

- Name
- Employee ID
- Position
- Username
- Password
- Confirm Password
- Email
- Department

i. If the registration is successful, the student should be redirected to the **Login** page. (Refer *Figure 3*)

ii. Use form validation techniques on the registration form to implement the requirements given below (either client-side or server-side):

- Every field must be completed; none can be left blank.
- The name and username fields permit only letters and spaces; other characters are not accepted.
- The email field should be a valid format.
- The password must consist of 8 to 20 characters and include at least one uppercase letter along with one special character.

iii. If any of the above given (ii) constraints are violated, you should display the suitable *error message* for the following cases:

- Submit the form with an empty field.
- Submit the form with numbers or/and special characters in the name and username field.
- Submit the form with an invalid email format.
- Submit the form without fulfilling the specified password validation criteria outlined in (ii).

(d) Create a login page as shown in *Figure 3*.

- i. If the login is successful, the Staff should be redirected to the inventory dashboard page (Refer *Figure 4*).
- ii. If the login is not successful, an error message should be displayed.

(e) Create the web pages to achieve the following tasks.

- i. Create an interface for *Add new item* and write a PHP Program to add new item record.
- ii. Write a PHP Program to *View all items'* details in a table and in the last column display link for the actions(Edit, delete).
- iii. Write PHP Program to *edit* the item detail if user click the EDIT button in the inventory table.
- iv. Write a PHP Program to *delete* the item detail if user click the Delete Button in the inventory table
- v. Write a php program to return *count* of computer tables.

(f) Write a JavaScript function that changes the background colour of the button when the "Click Me" button is clicked.

(g) Create a "logout" link for all the web pages except the registration page and login page.

[100%]



University of Vavuniya

Second Examination in Information Technology - 2023

First Semester - January/February 2025

IT2133 Advanced Web Programming (Theory)

Answer Four Questions Only

Time : Two hours

1. (a) List down *five* basic elements of web design. [05%]
(b) Briefly describe *three* web design characteristics. *simplicity, Grid based layout, load time, mobile friendly.* [15%]
(c) Discuss the anatomy of web page. *header, menu, Side bar, footer, Content* [20%]
(d) Explain what is URL and list down the elements with the suitable examples. *Scheme, host, Port number, Fragment, Path* [20%]
(e) Briefly describe the features of DNS. *Global distribution, Dynamicity, Scalability, Reliability, Loose coherence* [20%]
(f) Differentiate FQDN and PQDN with examples. [20%]
2. (a) Define what is a web browser. List down *three* examples. *Google chrome, Safari, edge.* [15%]
(b) Briefly explain domain delegation with an example. *Sub domain.* [15%]
(c) Briefly describe the process of forward DNS and reverse DNS. *name → IP IP → Name.* [20%]
(d) Describe the key areas of Computer Security. *Confidentiality, Availability, Authentication, Integrity.* [15%]
(e) State *five* ways to prevent cyber attacks. [15%]
(f) Write short notes on each of the following security attacks.

i. DNS Spoofing	iii. Man in the middle
ii. Denial of service	iv. SQL injection

 [20%]

3. (a) Define the term Content Management System(CMS) in your own words. [15%]
 (b) State *five* advantages of CMS. [15%]
 (c) List down *five* popular common CMS. [10%]
 (d) Discuss why PHP framework is commonly used for web application development in software industries. [20%]
 (e) List down *four* rules for PHP variables. [15%]
 (f) Discuss the *three* different variable scopes with suitable examples. [25%]

4. (a) Explain the syntax of an *if* statement in PHP with the aid of an example. [10%]
 (b) Write a PHP script to find the largest number among *three* numbers using nested if conditions. [15%]
 (c) State what is a constant. Explain with a suitable example. [15%]
 (d) Compare and contrast *GET* and *POST* in PHP. [15%]
 (e) Write a PHP script that uses a while loop to calculate the sum of all even numbers between 1 and 100. [25%]
 (f) Describe the differences between the functions *require()* and *include()* in PHP. [20%]

5. (a) Differentiate cookies and session. [15%]
 (b) What is the Document Object Model (DOM)? Describe its role in web development. [20%]
 (c) State *two* different ways of embedding JavaScript code in an HTML file. [20%]
 (d) What are the features of a good frame work. [15%]
 (e) Differentiate Library and framework. [15%]
 (f) List down the differences between HTML and XML. [15%]

Part III – Writing

Q.11 Write a **conversation** (minimum 15 exchanges) between two classmates about the book they read. (Spend 10 minutes)

(07 marks)

Q.12 Make a **paragraph** using the given topic sentence. (Spend 15 minutes)

"In today's fast-paced world, stress management techniques are essential for maintaining mental and emotional well-being."

(08 marks)

Q.13 Compose a **complaint letter** to a goods manufacturing or marketing company regarding a faulty product or to a service-providing company regarding an unsatisfactory service. Describe the issue in detail, explain how it has inconvenienced you, and suggest possible solutions or resolutions you would like to see. (Spend 15 minutes) Word limit: 200-250 words

(10 marks) 151

Q.14 Write an **essay** on the following topic incorporating the key areas mentioned below. (Not less than 300 words) (Spend 20 minutes)

Topic: The support of technology to solve the Sri Lankan Economic Crisis

- Manufacturing sector and distribution sector
- Marketing sector and Banking sector
- Public sector and private sector administrative or management or service systems
- Agricultural and Fisheries sector
- Educational and health sectors

(15 marks)

(Spend 10 minutes for complete revision)